



Sri Lanka Institute of Advanced Technological Education (SLIATE)

Advanced Technological Institute - Kurunegala

Higher National Diploma in Information Tech Batch-2324(FT)

Custom Bat & Ordering System for CS Bat

SRS

(Software Requirement Specification)

2nd Year – 2nd semester

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1. Introduction

1.1 Purpose of this Document

The primary purpose of this Software Requirement Specification (SRS) document is to clearly define the functional and technical requirements for the **Custom Bat & Ordering System** for CS Bat. This document provides a comprehensive blueprint for developers and stakeholders, outlining the expected behavior, features, and constraints of the platform.

CS Bat currently operates as a traditional physical storefront, which limits its customer base to local players and creates challenges in managing custom equipment requests. This digital solution aims to bridge the geographical gap and solve the technical "knowledge gap" amateur cricketers face when selecting gear. By defining requirements like custom order tracking, BMI-based recommendations, and automated SMS alerts, this SRS ensures the final system meets the operational needs of both the owner and the players.

The system focuses on achieving the following key objectives:

- **Defining System Functionality:** To describe all features such as student/user registration, BMI-based bat selection, and custom order tracking.
- **Guiding the Development Process:** To act as a comprehensive guide for the developer to ensure all modules are built according to specifications.
- **Establishing a Baseline for Testing:** To serve as a checklist for the testing phase to ensure every feature works before deployment.

2. General Description

2.1 Objective

The primary objective of the CS Bat system is to digitize and automate the custom cricket bat business. The system focuses on three main goals:

- **Improving Accessibility:** Providing a 24/7 e-commerce platform that allows customers across the country to browse and purchase bats.
- **Expert Guidance:** Implementing a BMI-based algorithm to suggest ideal bat weight and height based on a player's physical metrics.
- **Process Automation:** Replacing manual record-keeping with a centralized dashboard for inventory, production tracking, and profit analysis.

2.2 User Characteristics

1. **Administrator (Owner):** The authority who manages the product catalog, monitors stock levels, updates order statuses, and reviews monthly profit reports.
2. **Customers (Players):** Users who register to create profiles, use the smart recommendation tool, place custom orders, and track their delivery status.

2.3 Features and Benefits

1. **Mass Customization:** Players can specify edge size, weight, handle type, and colors for a personalized playing experience.
2. **Production Efficiency:** The system automates a 2-day production timeline and generates daily production lists for the workshop.
3. **Real-time Communication:** Automated SMS notifications via Twilio API keep customers informed of every status change.

3. Functional Requirements

3.1 Product and Catalog Management

- The system shall allow the Admin to manage the digital catalog, including adding wood types, bat models, and stock levels.
- The system shall provide filtering capabilities so users can find bats by material, price range, or softball/hardball categories.

3.2 Smart Recommendation and Customization

- **BMI Logic:** The system shall apply an algorithm to the user's height and weight to suggest the ideal bat dimensions.
- **Custom Interface:** The system shall feature a dedicated interface for users to select precise weight, handle types, grip colors, and sticker designs.

3.3 Order and Production Tracking

- **Automated Timeline:** Upon order confirmation, the system shall automatically trigger a 2-day production countdown.
- **Status Management:** The Admin shall update statuses (Confirmed, In Production, Ready, Shipped), which will be visible on the customer's private profile.

3.4 Financial and Data Management

- The system shall aggregate sales data to generate monthly profit and loss reports.
- The system shall track inventory in real-time, notifying the admin when raw materials or stock are low.

4. Interface Requirements

4.1 User Interfaces (UI)

- **Customer Portal:** A clean, responsive UI built with React.js that allows for intuitive bat configuration and easy navigation through the store.
- **Admin Dashboard:** A centralized control panel featuring visual charts for sales trends and structured lists for pending orders and delivery schedules.

4.2 Software Interfaces

- **Twilio SMS API:** For dispatching automated, real-time order status notifications to mobile numbers.
- **Database Interface:** Interaction with a MongoDB database for storing relational data like user profiles and order details.
- **Payment Gateway:** Integration of a digital payment flow to handle immediate transactions.

5.Design Constraints

- **Technology Stack:** Strictly restricted to Node.js, Express, React.js, and MongoDB.
- **Connectivity:** Requires a stable internet connection for cloud database access and API testing.
- **Data Integrity:** SMS delivery is dependent on the accuracy of the Sri Lankan mobile numbers provided.
- **Production Limit:** The system is built around a fixed 2-day production timeline logic.

6. Non-Functional Attributes

6.1 Security

- **Authentication:** Secure login and session management for both customers and the administrator.
- **Data Integrity:** Ensuring that order details and payment records are protected from unauthorized access.

6.2 Performance

- **Notification Speed:** SMS updates via the Twilio API should be dispatched within seconds of an admin update.
- **Responsiveness:** The web application must be optimized for fast loading across different devices.

6.3 Maintainability

- **Modular Design:** The system shall use React's component-based architecture to allow for easy updates and future feature additions.

7. Appendices

7.1 Glossary of Terms and Abbreviations

- **SRS:** Software Requirement Specification.
- **BMI:** Body Mass Index (used for recommendation algorithms).
- **Twilio:** A cloud communications platform for SMS.
- **MERN Stack:** MongoDB, Express, React, Node.js (technologies used for development).

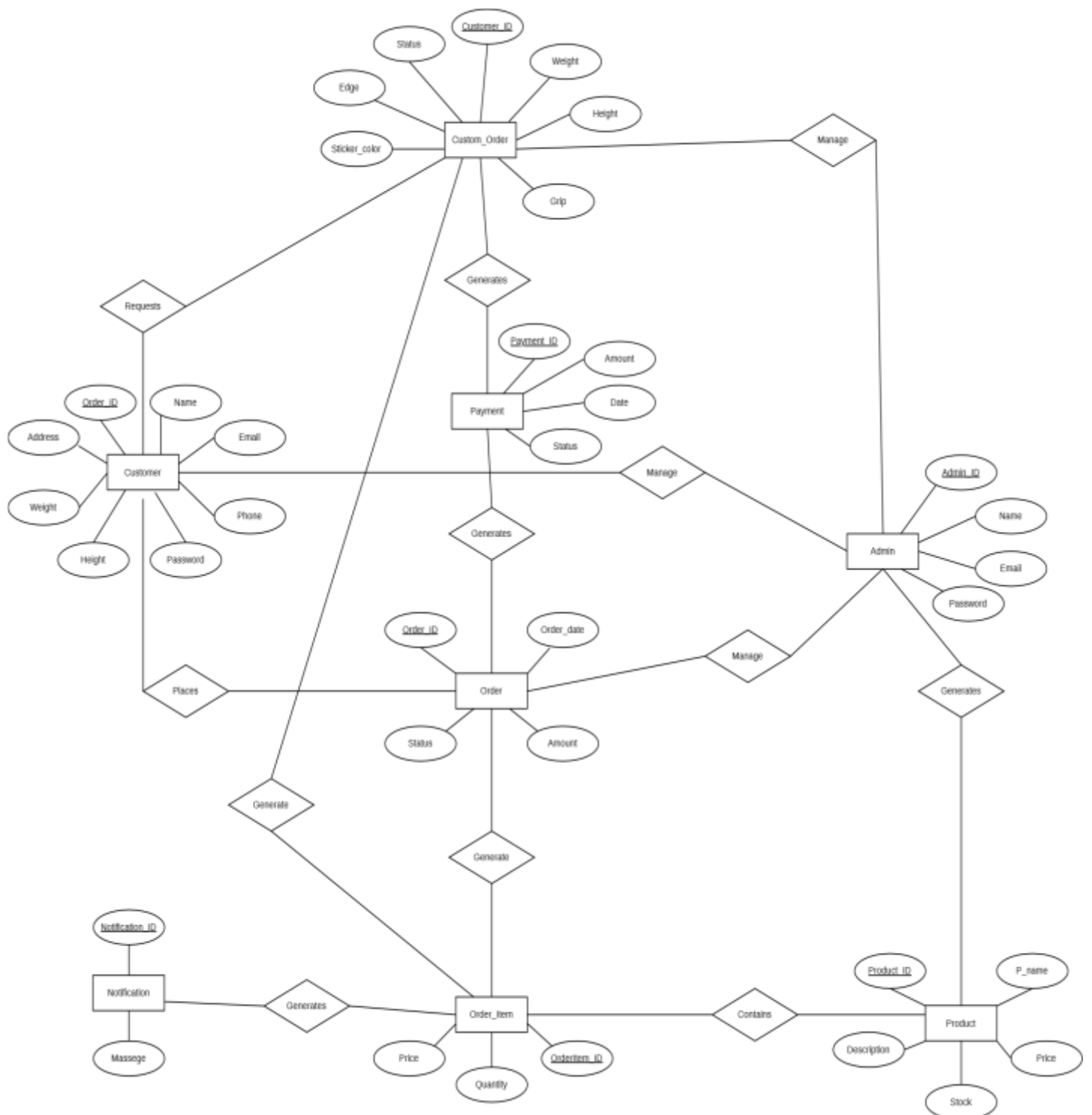
7.2 Hardware and Software Requirements

- **Hardware:** Intel Core i5 processor, 8GB RAM, and 256GB SSD (Recommended).
- **Software:** Node.js (Backend), MongoDB (Database), React.js (Frontend), VS Code, and Postman.

7.5 References

- **React Official Documentation.** *Getting Started with React and Hooks.* Available at: <https://react.dev/> (Accessed: 12 April 2026).
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ER Diagram



Use case

CS Custom Bat Ordering System

